

GRACE: A Narrative-Centered Socio-Technical Ecosystem for Elder Wellbeing

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Abstract—Population aging is increasing demand for digital systems that support older adults’ wellbeing. Most existing systems emphasize safety, monitoring, and service coordination, but underrepresent identity, meaning, contribution, and relationship quality. This paper proposes GRACE (Growing, Resourceful, Active, and Enriched Life), a narrative-centered socio-technical ecosystem for elder wellbeing. GRACE is designed as an intelligent multiagent community computing framework that supports older adults in maintaining relationships with the spiritual self, family and friends, working colleagues, communities, and the wider world, including markets and public services. The paper makes four contributions. First, it introduces a narrative-centered theory of elder wellbeing grounded in relationship quality and contribution exchange. Second, it proposes the GRACE Index, a daily measure of whether a person remains in the zone of narrative selfhood during ordinary and difficult periods. Third, it extends the framework with three innovations: daily hero’s-journey story creation, an AI-based fast state predictor, and a smart stimulus agent for anticipatory support. Fourth, it presents an agent-based simulation design to evaluate these innovations. We argue that elder-oriented AI should move beyond deficit-oriented monitoring toward socio-technical ecosystems that preserve dignity, agency, reciprocity, and life authorship.

Index Terms—aging, wellbeing, narrative identity, multiagent systems, human-centered AI, socio-technical systems, community computing, eldercare technology

I. INTRODUCTION

POPULATION aging is reshaping healthcare, labor, family structures, and public services. In response, digital systems for older adults have grown rapidly, especially in telehealth, remote monitoring, medication support, and care coordination. These systems are valuable, but many are built around narrow assumptions: that old age is primarily a condition of decline, risk, and dependency.

That assumption is socially and technically limiting. Older adults are not only patients or service users. They are also storytellers, mentors, workers, caregivers, citizens, spiritual beings, and bearers of memory and legacy. Their wellbeing depends not only on safety and functional support, but also on continuity of identity, meaningful relationships, opportunities to contribute, and the capacity to interpret life events within a coherent story.

This paper proposes GRACE—Growing, Resourceful, Active, and Enriched Life—as a socio-technical ecosystem for elder wellbeing. GRACE is an intelligent community computing

framework organized around five relational domains: 1) spiritual self, 2) family and friends, 3) working colleagues, 4) communities, and 5) the wider world, including markets and public services. GRACE is grounded in the premise that elder wellbeing is relational and that the quality of relationships shapes the exchange of care, recognition, participation, and contribution.

The central design claim of GRACE is that narrative identity should be treated as a core systems concept. Older adults should not be modeled merely through static profiles, health states, or risk scores. They should be understood as persons living through unfolding stories. GRACE therefore acts as a live autobiographical ecosystem, helping elders remain connected to selfhood across daily life, transitions, and adversity. A high-level overview is shown in Fig. 1, with the multiagent architecture shown in Fig. 2 and the dynamic closed-loop logic in Fig. 3.

To operationalize this vision, the paper introduces the GRACE Index, a daily measure of whether a person remains “in the zone” of narrative selfhood. The paper further adds three innovations: 1) a Daily Story Creation Engine that frames lived experience through the hero’s journey, 2) an AI-based Fast State Predictor that forecasts near-future wellbeing state, and 3) a Smart Stimulus Agent that initiates timely, personalized prompts to preserve wellbeing and authorship. Finally, the paper proposes an agent-based simulation design to demonstrate these innovations.

A. Contributions

1. A socio-technical framework for elder wellbeing centered on narrative identity, relationship quality, and contribution exchange.
2. The GRACE Index, a daily metric of narrative selfhood and adaptive continuity.
3. Three AI-enabled innovations—daily story creation, fast state prediction, and smart stimuli—for anticipatory wellbeing support.
4. An agent-based simulation design for evaluating these mechanisms before real-world deployment.

II. RELATED WORK

A. Aging Technologies and Deficit-Oriented Design

Technologies for older adults have often focused on safety, surveillance, and functional support (Sixsmith & Gutman, 2013; Czaja et al., 2006; Mitzner et al., 2010). While these systems can improve independence and care delivery, they may also reproduce deficit-oriented assumptions about aging. HCI scholarship has argued that such framings narrow the design

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space and underplay agency, participation, and dignity (Vines et al., 2015).

B. B. Social Connectedness and Wellbeing

Social relationships are strongly associated with wellbeing, resilience, and mortality outcomes (Holt-Lunstad et al., 2010). Social isolation and perceived loneliness can negatively affect cognition and health (Cacioppo & Hawkley, 2009). In later life, relationship quality matters as much as contact frequency, including dimensions such as reciprocity, belonging, and social integration (Cornwell & Waite, 2009; Berkman et al., 2000).

C. C. Narrative Identity and Later Life

Narrative identity theory suggests that individuals construct a sense of self through evolving life stories (McAdams, 2001; Habermas & Bluck, 2000; Singer, 2004). In later life, autobiographical continuity and life review become especially important for meaning-making, legacy, and coping with transition (Butler, 1963; Bohlmeijer et al., 2007). Yet most digital systems do not operationalize narrative identity as a core computational construct.

D. D. Generativity and Productive Aging

Older adults continue to contribute through mentoring, caregiving, volunteering, advising, and cultural transmission. Generativity is a major source of meaning and self-worth (McAdams & de St. Aubin, 1992; Erikson, 1982). Productive aging research likewise emphasizes the value of creating roles and systems that support continued contribution (Morrow-Howell et al., 2017).

E. E. Community Computing, Multiagent Systems, and Human-Centered AI

Community informatics has shown how digital infrastructures can support participation, local engagement, and social inclusion (Gurstein, 2007; Carroll, 2014). Multiagent systems offer a framework for distributed intelligence, coordination, and adaptive decision-making (Jennings et al., 1998; Wooldridge, 2009). Human-centered AI emphasizes reliability, safety, and alignment with human values (Shneiderman, 2020). GRACE integrates these strands into a single socio-technical architecture.

F. F. Research Gap

Existing work rarely integrates all of the following in one coherent framework: narrative identity, relationship quality, contribution exchange, community participation, institutional and market access, predictive wellbeing support. GRACE addresses this gap by proposing a narrative-centered, socially grounded AI ecosystem for elder wellbeing.

III. GRACE AS A SOCIO-TECHNICAL ECOSYSTEM

The overall socio-technical structure of GRACE is illustrated in Fig. 1. The five relational domains, their functions, and example system supports are summarized in Table I.

A. A. Core Premise

GRACE is based on a simple claim: elder wellbeing emerges from the interaction of identity, relationships, contribution, and access. A person flourishes when they remain connected to who they are, to people who matter, to ways of contributing, and to the broader systems that structure everyday life.

B. B. Five Relational Domains

GRACE supports the elder's relationship with: 1. Spiritual self 2. Family and friends 3. Working colleagues 4. Communities 5. Wider world, including markets and public services.

Table I: Main Caption
(a) First Table

Domain	Description
Spiritual Self	Inner life, values, faith, reflection, existential orientation
Family and Friends	Intimate and enduring personal relationships
Working Colleagues	Ongoing or legacy vocational and professional ties
Communities	Local groups, associations, faith communities, neighborhoods, civic
Wider World	Markets, institutions, services, transport, civic systems, public serv

(b) Second Table

Col1	Col2	Col3
A	B	C
E	F	G
A	G	G

Table II: A nicely wrapped table using kableExtra

Doman	Description	Functions	Examples
Very long introductory sentence that will wrap	Another long explanation for a completely different scenario.	Very long introductory sentence that will wrap	Another long explanation for a completely different scenario.
Short text	Simple detail	Short text	Simple detail

Table III: Domain, description, wellbuing functions, adn examples system supports.

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Table IV: Main Caption

(a) First Table

Col1	Col2	Col3
A	B	C
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(b) Second Table

Col1	Col2	Col3
A	B	C
E	F	G
A	G	G

IV. SAMPLES

file is intended to serve as a "sample article file" for IEEE journal papers produced with (Pandoc/Quarto)-Markdown using IEEEtran.cls version 1.8b and later for the PDF output. It is based on bare_jrnl_new_sample4.tex provided by IEEE Publication Technology, Staff and available from

<https://template-selector.ieee.org/>. The most common elements are covered in the simplified and updated instructions in `New_IEEEtran_how-to.pdf`. For less common elements you can refer back to the original `IEEEtran_HOWTO.pdf`. It is assumed that the reader has a basic working knowledge of L^AT_EX [1] and of (Pandoc/Quarto)-Markdown [2], [3] markup.

1) *Figures*: An image with nonempty alt text will be rendered as a figure with a caption with Pandoc and Quarto. Quarto includes a different features to simplify the use of figures and subfigures. Here, it is recommended to use `div` block with `#fig-` label to embed your Figures.



Fig. 1: An example of figure.



(a) (b)
Fig. 2: An example with sub-figure.

The figures is cross-referenced as Fig. 2 and even the sub-figures as Fig. 2b.

2) *Tables*: Similarly, many kind of tables may be used with Pandoc and Quarto. The latter also includes different features to simplify the table output. To make tables cross-referenceable use a label with a `#tbl-` prefix.

However, it is recommended to avoid using the commonly used single Markdown table known as a ‘pipe table’. In fact, Pandoc Markdown uses the L^AT_EX `longtable` package, which does not support the two-column mode, which is required for most IEEEtran journals. `quarto-ieee` uses a hack to temporarily switch to one-column mode. However, this hack may break the page layout. To overcome this issue, a basic way is to use code cells (as for Table VI). Quarto is a multi-language and it uses `Knitr` to execute R code and can execute Python code blocks within Markdown.

Table V: Main Caption

(a) First Table

Col1	Col2	Col3
A	B	C
E	F	G
A	G	G

(b) Second Table

Col1	Col2	Col3
A	B	C
E	F	G
A	G	G

The Tables are cross-referenced as Table V for details, especially Table Vb. There is also Table VI.

Table VI: A table

Col1	Col2	Col3
A	D	G
B	E	H
C	F	I

A. Bibliography

IEEE journal should normally use IEEEtran¹ B_IT_EX style. Nevertheless, Pandoc and Quarto do support B_IT_EX with `natbib` or `biblatex`. However, neither is officially recommended for normal IEEE use. For this reason, `quarto-ieee` uses `citeproc` with the `ieee` CSL style sheet.

V. CONCLUSIONS

The conclusion goes here.

ACKNOWLEDGMENT

This should be a simple paragraph before the References to thank those individuals and institutions who have supported your work on this article.

APPENDIX AN APPENDIX

Use `[] {.appendix options="An Appendix"}` markup if you have a single appendix. IEEEtran state that to do not use `\section{}` anymore after `\appendix`.

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First Author Use `IEEEbiography` with figure as option and the author name as the argument followed by the biography text.

Second Author Use `IEEEbiographynophoto` and the author name as the argument followed by the biography text.

¹IEEEtran BibT_EX style support page is: <http://www.michaelshell.org/tex/ieee/ieee-tran/bibtex/>